

## Exp No: 14

### EMBEDDED C PROGRAM TO DISPLAY “HELLO WORLD” MESSAGE

#### OBJECTIVES

- To write an Embedded C program to display the “Hello World” message using the AT89C51 microcontroller.
- To configure the 8051 UART serial communication for transmitting the message.
- To verify the transmitted “Hello World” message using the Keil simulation environment.

#### PROGRAM

```
#include <reg51.h> #include <stdio.h>

void main (void){

SCON = 0x50; /* SCON: mode 1, 8-bit UART, enable rcvr */ TMOD = 0x20;

/* TMOD: timer 1, mode 2, 8-bit reload */ TH1 = 0xFD; /* TH1: reload

value for 9600 baudrate */

TR1 = 1; /* TR1: timer 1 run */

TI = 1; /* TI: set TI to send first char of UART */ while (1)
{
printf ("Hello World ! \n ");
}
}
```

# OUTPUT

The screenshot displays the uVision IDE interface for a project named 'HELLO\_WORLD'. The main window shows the disassembly of the program, with the source code visible in the background. The registers window on the left shows the current state of the registers, and the command window at the bottom shows the output of the program.

**Registers Window:**

| Register | Value      |
|----------|------------|
| R0       | 0x00       |
| R1       | 0x37       |
| R2       | 0x00       |
| R3       | 0x00       |
| R4       | 0x00       |
| R5       | 0x00       |
| R6       | 0x00       |
| R7       | 0x00       |
| R8       | 0x00       |
| R9       | 0x00       |
| R10      | 0x00       |
| R11      | 0x00       |
| R12      | 0x00       |
| R13      | 0x00       |
| R14      | 0x00       |
| R15      | 0x00       |
| PC       | 0x00000000 |
| SP       | 0x00000000 |
| SR       | 0x00000000 |
| SR_0     | 0x00000000 |
| SR_1     | 0x00000000 |
| SR_2     | 0x00000000 |
| SR_3     | 0x00000000 |
| SR_4     | 0x00000000 |
| SR_5     | 0x00000000 |
| SR_6     | 0x00000000 |
| SR_7     | 0x00000000 |
| SR_8     | 0x00000000 |
| SR_9     | 0x00000000 |
| SR_10    | 0x00000000 |
| SR_11    | 0x00000000 |
| SR_12    | 0x00000000 |
| SR_13    | 0x00000000 |
| SR_14    | 0x00000000 |
| SR_15    | 0x00000000 |

**Disassembly Window:**

```
9: {
10: printf ("Hello World ! \n ");
C:\0x0C1E 7BFF MOV R3,#0xFF
C:\0x0C20 7BFF MOV R2,#0xFF
C:\0x0C22 7BFF MOV R1,#0xFF
--
main.c STARTUP.AS1
1 #include <reg51.h> #include <stdio.h>
2 void main (void) {
3     SCON = 0x50; /* SCON: mode 1, 8-bit UART, enable rcvr */ TH0 = 0x20;
4     /* TH0: timer 1, mode 2, 8-bit reload */ TH1 = 0xFF; /* TH1: reload
5
6     -value for 9600 baudrate */
7     TR1 = 1; /* TR1: timer 1 run */
8     TI = 1; /* TI: set TI to send first char of UART */ while (1)
9     {
10    printf ("Hello World ! \n ");
11    }
12 }
```

**Command Window:**

```
Running with Code Size Limit: 2K
Load "C:\Users\M.POOJITH GANESH\OneDrive\embedded\Experiment 14\Objects\HELLO_WORLD"
```

**UART #1 Window:**

```
Hello World !
Hello World !
Hello World !
Hello World !
Hello World !
Hello World !
```

**Simulation Status:**

Simulation t1: 174.61861950 sec L10 C30 CAP. NUM. SCRL. OVR. R/W